

w : weights that are quantised

x : input.

$$\hat{w} = w + \Delta \quad \Delta : \text{error added to original vector}$$

What is $\langle x, \hat{w} \rangle$

and how well it approximates $\langle w, x \rangle$

In case of deterministic rounding to nearest,

assume that the precision is δ

$$\hat{w} = w + \Delta \quad (|\Delta[i]| \leq \frac{\delta}{2})$$

$$\langle x, \hat{w} \rangle$$

$$= \sum x_i w + x_i \Delta_i$$

$$= \langle x, w \rangle + x_i \Delta_i$$

Worst Case $x_i y_i > 0 \forall i, j$
 and error $\Rightarrow \Delta_i = \frac{\delta}{2} \forall i$

Then, in worst case

$$\langle x, \hat{w} \rangle = \langle x, w \rangle + \frac{\delta}{2} \sum x_i$$

$$\langle x, \hat{w} \rangle = \langle x, w \rangle + \frac{\delta}{2} \|x\|_1$$

Stochastic Rounding

$$\hat{w} = w + \Delta$$

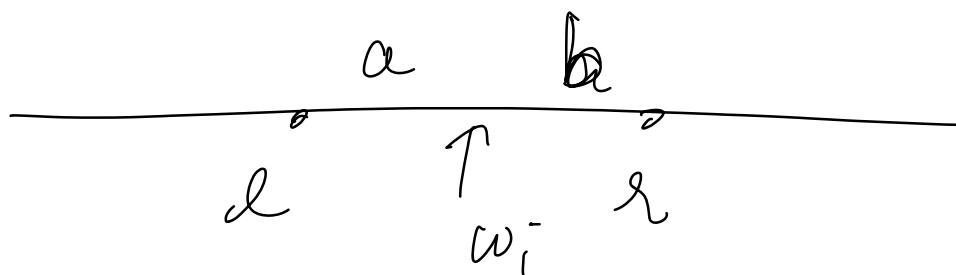
$$\Delta_i =$$

with probability

$$\frac{b}{a+b}$$

$$+ b$$

$$\frac{a}{a+b}$$



$$E(\Delta_i) = 0$$

$$E(\hat{\omega}) = \omega$$

$$\begin{aligned} \langle x, \hat{\omega} \rangle &= \sum x_i (\omega + \Delta_i) \\ &= \langle x, \omega \rangle + \sum x_i \Delta_i \end{aligned}$$

$$E(\langle x, \hat{\omega} \rangle) = \langle x, \omega \rangle$$

$$\text{Var}(\langle x, \hat{\omega} \rangle) = \sum x_i^2 \cdot \text{Var}(\Delta_i)$$

$$= \sum x_i^2 \sigma^2$$

$$= \sigma^2 \|x\|_2^2$$

$$= \frac{\sigma^2}{4} \|x\|_2^2$$

$$\text{MSE}_{\text{nearest rounding}} = \frac{\sigma^2}{4} \|x\|_1^2$$

$$MSE_{\text{stochastic rounding}} = \frac{\delta^2}{4} \|x\|_2^2$$

① $MSE_{\text{nearest rounding}}$ can be smaller based on actual x

Additional Reading $\Rightarrow$ Randomness of x matters
 why universal hash functions work - Mitchen Mader.

② Worst Case.

$MSE_{\text{round to nearest}}$

$> MSE_{\text{stochastic nearest}}$